HTML Concepts Learned Today

Beginner Theory Notes — Personal Profile Page / Web Page Fundamentals

1. Introduction to HTML and Web Page Structure

• HTML stands for **HyperText Markup Language**. It is the standard markup language used to create and structure content on web pages.

• HTML describes what content is: headings, paragraphs, images, links, lists, and other page elements. HTML is a markup language, not a programming language.

• **HTML** provides structure, **CSS** controls appearance and styling, and **JavaScript** adds behavior and interactivity.

• Web browsers such as Chrome, Edge, Firefox, and Safari read HTML and render the document as a visual web page.

2. Basic HTML Document Structure

• **<!DOCTYPE html>** — declares that the document uses HTML5.

• **<html>** — the root element containing the complete HTML document.

• **<head>** — contains information about the document, such as its title, metadata, and references to CSS or scripts.

• **<title>** — specifies the title shown in the browser tab.

• **<body>** — contains the visible content of the web page.

3. Tags and Elements

• An HTML **tag** is written using angle brackets, such as .

• An **element** generally consists of an opening tag, content, and a closing tag. Example conceptually: Content .

• Most elements have opening and closing tags. Some are **void elements** and do not require a closing tag, such as the image and line-break elements.

4. Headings

• HTML provides six heading levels: **h1 through h6**.

• <h1> represents the main/highest-level heading, while <h6> represents the lowest-level heading.

• Headings should describe the hierarchy of content, not simply be chosen because they look larger.

5. Paragraphs

• The **p** element represents a paragraph of text.

• Separate paragraphs make written content easier to organize and read.

6. Comments

• HTML comments are notes written in the source code that are not displayed as page content.

• They are useful for explaining code, labeling sections, leaving notes, and temporarily disabling code during development.

• Comment syntax: **<!-- comment -->**

7. Links

• The **a (anchor)** element creates hyperlinks.

• The **href** attribute specifies the destination of a link.

• The **target** attribute can be used to control where a link opens; for example, _blank is commonly used to open it in a new tab.

8. Images

• The **img** element is used to display images.

• The **src** attribute specifies the image source or location.

• The **alt** attribute provides alternative text describing the image. It is important for accessibility and is useful when an image cannot be displayed.

9. Lists

• **Unordered list (<ul>)** — displays items as a bulleted list.

• **Ordered list (<ol>)** — displays items in a numbered/ordered list.

• **List item (<li>)** — represents an individual item inside an ordered or unordered list.

10. Text Formatting

• **** represents strong importance and is commonly displayed in bold.

• **<b>** makes text bold without necessarily expressing semantic importance.

• **** represents emphasis and is commonly displayed in italic text.

• **<i>** represents text in an alternate voice or mood and is commonly displayed in italic.

• **<u>** represents underlined text.

• **
** creates a line break.

• **<mark>** represents highlighted/relevant text.

11. Attributes

• Attributes provide additional information or configuration for an HTML element.

• They are generally written inside the opening tag as a name and value.

• Examples you learned include **href** for links, **src** for images, **alt** for alternative image text, and **target** for link behavior.

12. Key Differences to Remember

• **HTML vs CSS:** HTML defines structure/content; CSS defines appearance and styling.

• **<head> vs <body>:** head contains document information and resources; body contains visible page content.

• ** vs <ol>:** ul is unordered/bulleted; ol is ordered/numbered.

• **Tag vs element:** a tag is the markup itself; an element generally includes the opening tag, content, and closing tag.

13. Personal Profile Page — Applying the Concepts

• For a beginner personal profile page, useful sections include: your name and role, About Me, Education, Skills, Interests, Profile Image, Useful Links such as LinkedIn/GitHub, and a simple Contact section.

• Try to demonstrate the concepts you learned: headings for sections, paragraphs for descriptions, comments to organize your HTML, links, an image, ordered/unordered lists, and basic text formatting.

• The goal is to understand why each HTML element is being used rather than simply making the page visually attractive.

Quick Revision Checklist

• HTML = HyperText Markup Language

• DOCTYPE declares HTML5

• html = root element

• head = document information/resources

• body = visible content

• h1–h6 = headings

• p = paragraph

• a + href = link

• img + src + alt = image

• ul/ol + li = lists

• strong/em = semantic emphasis

• Comments are not displayed on the page

• Attributes provide additional information about elements

Tip: Build your assigned profile page yourself and make sure you can explain every HTML element you use.